

Factoring Cubes and Higher-Degree Expressions

Answer Key

Algebra 2

Quick Answers

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|-----------------------------------|--|
| 1. $(x - 2)(x^2 + 2x + 4)$ | 6. -3, -2, 3 |
| 2. $(3x + 4)(9x^2 - 12x + 16)$ | 7. -2, -1, 1, 2 |
| 3. $(2x - 6)(x^2 + 3x + 9)$ | 8. $(2x^2 - 3)(4x^4 + 6x^2 + 9)$ |
| 4. $(x - 2)(x + 2)(x + 3)$ | 9. -2, 2, 4 |
| 5. $(x - 3)(x - 2)(x + 2)(x + 3)$ | 10. $3x^2(x - 2)(x^2 + 2x + 4)$, 0, 2 |
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Curriculum

Level: Algebra 2

HSA-APR.B.3 – problems 6, 7, 9, 10

HSA-SSE.B.3 – problems 1, 2, 3, 4, 5, 8

Difficulty 1–5 of 5 across 12 tagged checks.

1. Difference of cubes: $x^3 - 8$. Both terms are perfect cubes: $x^3 = (x)^3$ and $8 = 2^3$, so $a = x$ and $b = 2$. The difference pattern is $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. Substituting: $a - b = x - 2$, and $a^2 + ab + b^2 = x^2 + 2x + 4$. The trinomial does not factor further — its discriminant is $4 - 16 = -12 < 0$.

$$(x - 2)(x^2 + 2x + 4)$$

2. Sum of cubes: $27x^3 + 64$. Write each term as a cube: $27x^3 = (3x)^3$ and $64 = 4^3$, so $a = 3x$, $b = 4$. The sum pattern is $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ — note the middle sign flips. Substituting: $a + b = 3x + 4$, $a^2 = 9x^2$, $ab = 12x$, $b^2 = 16$. Expanding back gives $27x^3 - 36x^2 + 48x + 36x^2 - 48x + 64 = 27x^3 + 64$, which confirms the middle terms cancel.

$$(3x + 4)(9x^2 - 12x + 16)$$

3. Common factor first: $2x^3 - 54$. Neither $2x^3$ nor 54 is a perfect cube, so the pattern cannot be applied yet. The greatest common factor is 2: $2x^3 - 54 = 2(x^3 - 27)$. Now the bracket is a difference of cubes with $a = x$ and $b = 3$: $x^3 - 27 = (x - 3)(x^2 + 3x + 9)$. Skipping the GCF is the usual way this problem goes wrong — $2x^3$ has no integer cube root.

$$2(x - 3)(x^2 + 3x + 9)$$

4. Factor by grouping: $x^3 + 3x^2 - 4x - 12$. Group in pairs: $(x^3 + 3x^2) + (-4x - 12)$. Pull x^2 from the first pair and -4 from the second: $x^2(x + 3) - 4(x + 3)$. The matching binomial $(x + 3)$ is the signal that the grouping worked, so factor it out: $(x + 3)(x^2 - 4)$. The second factor is a difference of squares and must still be broken up: $x^2 - 4 = (x - 2)(x + 2)$.

$$(x + 3)(x - 2)(x + 2)$$

5. Quadratic in disguise: $x^4 - 13x^2 + 36$. The exponents are 4 and 2, so with $u = x^2$ the expression is $u^2 - 13u + 36$. Two numbers multiplying to 36 and adding to -13 are -4 and -9 : $u^2 - 13u + 36 = (u - 4)(u - 9)$. Substituting back: $(x^2 - 4)(x^2 - 9)$. Each bracket is a difference of squares, so the factorisation is not finished until both are split: $x^2 - 4 = (x - 2)(x + 2)$ and $x^2 - 9 = (x - 3)(x + 3)$.

$$(x - 2)(x + 2)(x - 3)(x + 3)$$

6. From factors to zeros: $P(x) = x^3 + 2x^2 - 9x - 18$. Group: $(x^3 + 2x^2) + (-9x - 18) = x^2(x + 2) - 9(x + 2) = (x + 2)(x^2 - 9)$, and $x^2 - 9 = (x - 3)(x + 3)$, so $P(x) = (x + 2)(x - 3)(x + 3)$. Set each factor to zero: $x + 2 = 0$, $x - 3 = 0$, $x + 3 = 0$. In increasing order the real zeros are $x = -3, -2, 3$.

Sign table. Every zero is simple (multiplicity one), so P changes sign at each of them, and the leading term x^3 makes P positive far to the right. Reading right to left and alternating: $x < -3$ gives $-$; $-3 < x < -2$ gives $+$; $-2 < x < 3$ gives $-$; $x > 3$ gives $+$. Spot check at $x = 0$: $P(0) = -18$, negative, matching the third interval.

7. Solve a quartic: $x^4 - 5x^2 + 4 = 0$. With $u = x^2$ the equation is $u^2 - 5u + 4 = 0$, which factors as $(u - 1)(u - 4) = 0$. Substituting back: $(x^2 - 1)(x^2 - 4) = 0$, and each bracket is a difference of squares, so $(x - 1)(x + 1)(x - 2)(x + 2) = 0$. The zero-product property gives one solution per linear factor, and all four are real. In increasing order: $x = -2, -1, 1, 2$.

8. Find the mistake: $8x^6 - 27$. Sam identified the cubes correctly — $8x^6 = (2x^2)^3$ and $27 = 3^3$, so $a = 2x^2$ and $b = 3$ — and his binomial factor $2x^2 - 3$ is right. His error is in the trinomial. For a *difference* of cubes the pattern is $(a - b)(a^2 + ab + b^2)$: the middle term is $+ab$, not $-ab$. Sam flipped it, writing $-6x^2$ where the pattern requires $+6x^2$. (The quick test: expanding his version leaves the x^4 and x^2 terms uncanceled, so it cannot equal a two-term difference.) With $a^2 = 4x^4$, $ab = 6x^2$ and $b^2 = 9$ the correct factorisation is

$$(2x^2 - 3)(4x^4 + 6x^2 + 9)$$

Neither factor breaks up further over the integers: $2x^2 - 3$ has irrational roots, and $4x^4 + 6x^2 + 9$ is positive for every real x .

9. Zeros and shape: $Q(x) = x^3 - 4x^2 - 4x + 16$. Group: $(x^3 - 4x^2) + (-4x + 16) = x^2(x-4) - 4(x-4) = (x-4)(x^2-4)$, and $x^2-4 = (x-2)(x+2)$, so $Q(x) = (x-4)(x-2)(x+2)$. Setting each factor to zero and ordering the results, the real zeros are $x = -2, 2, 4$.

Model sketch (open response — compare the student's curve to this). The degree is odd with a positive leading coefficient, so the curve rises to the right and falls to the left. All three zeros are simple, so the graph crosses the axis at each one. That fixes the shape completely: below the axis for $x < -2$, above it between -2 and 2 (check $Q(0) = 16$), below it between 2 and 4 (check $Q(3) = -5$), and above it for $x > 4$. There is a local maximum between -2 and 2 and a local minimum between 2 and 4 ; exact turning points were not required. Accept any sketch with those four sign regions, three crossings, and that end behaviour.

10. Everything at once: $R(x) = 3x^5 - 24x^2$. The greatest common factor is $3x^2$: $3x^5 - 24x^2 = 3x^2(x^3 - 8)$. The bracket is now a difference of cubes with $a = x$, $b = 2$, giving $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$:

$$3x^2(x - 2)(x^2 + 2x + 4)$$

Real zeros. The factor $3x^2$ gives $x = 0$ (a double zero, so the graph touches the axis there rather than crossing), and $x - 2$ gives $x = 2$. In increasing order the real zeros are $x = 0, 2$.

The factor with no real zero is $x^2 + 2x + 4$. Its discriminant is $2^2 - 4(1)(4) = 4 - 16 = -12$, which is negative, so it never reaches the axis — completing the square shows the same thing, since $x^2 + 2x + 4 = (x + 1)^2 + 3 \geq 3$.